

Area Between Curves

ANSWER KEY



Area between two curves

- 1 Find the area of the region bounded by $y = x^2$ and $y = x + 2$.
Intersections: $x^2 = x + 2$, so $x^2 - x - 2 = 0$, $(x - 2)(x + 1) = 0$: $x = -1, 2$. Area
$$= \int_{-1}^2 [(x + 2) - x^2] dx = \left[\frac{x^2}{2} + 2x - \frac{x^3}{3} \right]_{-1}^2 = \frac{9}{2}.$$
- 2 The region bounded by $y = \sqrt{x}$, $y = 0$, and $x = 4$ is shown. Find its area.
$$\int_0^4 \sqrt{x} dx = \left[\frac{2}{3} x^{3/2} \right]_0^4 = \frac{16}{3}.$$

Setting up with respect to y , and total area

- 3 Find the area enclosed by $x = y^2$ and $x = 2y$ by integrating with respect to y .
Intersections: $y^2 = 2y$ gives $y = 0, 2$. Area $= \int_0^2 (2y - y^2) dy = \left[y^2 - \frac{y^3}{3} \right]_0^2 = 4 - \frac{8}{3} = \frac{4}{3}.$
- 4 Find the total area between $f(x) = \sin x$ and the x -axis on $[0, 2\pi]$ (note the curve is below the axis on part of this interval).
By symmetry, $\int_0^\pi \sin x dx = 2$ and $\left| \int_\pi^{2\pi} \sin x dx \right| = 2$. Total area $= 2 + 2 = 4$.